

CS 121 Assignment 3: Milestone 2

Search Component Report

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For M2 we built the search component on top of the inverted index from M1. To avoid loading the whole index into memory, we built a seek index that maps each token to its byte offset in the index file, so we can jump directly to any token when handling a query. Query terms get stemmed using the Porter stemmer, then we do a Boolean AND across all terms, score the matching docs with tf-idf, and return the top 5 results.

Query 1. cristina lopes

Rank	URL
1	https://www.ics.uci.edu/community/news/notes/notes_2010.php
2	https://www.ics.uci.edu/~hsajnani/
3	https://www.informatics.uci.edu/explore/faculty-profiles/cristina-lopes/
4	https://www.ics.uci.edu/faculty/profiles/view_faculty.php?ucinetid=lopes
5	https://www.ics.uci.edu/community/news/notes/notes_2011.php

Query 2. machine learning

Rank	URL
1	https://www.ics.uci.edu/~welling/publications/publications.html
2	https://cml.ics.uci.edu/category/aiml/
3	https://cml.ics.uci.edu/category/aiml/page/2/
4	http://mondego.ics.uci.edu/datasets/maven-contents.txt
5	https://www.ics.uci.edu/~pazzani/Publications/OldPublications.html

Query 3. ACM

Rank	URL
1	https://www.ics.uci.edu/~eppstein/pubs/pubs.ff
2	https://www.ics.uci.edu/~gmark/Home_page/Publications.html
3	https://www.ics.uci.edu/~eppstein/pubs/all.html
4	https://www.ics.uci.edu/~kay/courses/131/s04readings.html
5	https://www.ics.uci.edu/~eppstein/bibs/eppstein.html

Query 4. master of software engineering

Rank	URL
1	https://www.ics.uci.edu/~kay/courses/h22/hw/DVD-random.txt
2	https://www.ics.uci.edu/~kay/courses/h22/hw/DVD.txt
3	https://www.informatics.uci.edu/grad/mswe/
4	https://www.ics.uci.edu/~nenovita/vita.html
5	https://mswe.ics.uci.edu/

Search Interface Screenshots

1. cristina lopes

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Enter query: cristina lopes

Top results for 'cristina lopes' (0.0ms):
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1. https://www.ics.uci.edu/community/news/notes/notes_2010.php
   Score: 49.2576
2. https://www.ics.uci.edu/~hsajnani/
   Score: 47.9548
3. https://www.informatics.uci.edu/explore/faculty-profiles/cristina-lopes/
   Score: 43.7161
4. https://www.ics.uci.edu/faculty/profiles/view_faculty.php?ucinetid=lopes
   Score: 39.9341
5. https://www.ics.uci.edu/community/news/notes/notes_2011.php
   Score: 39.1123

Enter query: 

```

2. machine learning

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Enter query: machine learning

Top results for 'machine learning' (18.0ms):
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1. https://www.ics.uci.edu/~welling/publications/publications.html
   Score: 26.4294
2. https://cml.ics.uci.edu/category/aiml/
   Score: 24.1462
3. https://cml.ics.uci.edu/category/aiml/page/2/
   Score: 24.0817
4. http://mondego.ics.uci.edu/datasets/maven-contents.txt
   Score: 23.6419
5. https://www.ics.uci.edu/~pazzani/Publications/OldPublications.html
   Score: 21.5025

Enter query: 

```

3. ACM

Enter query: ACM

Top results for 'ACM' (8.0ms):

- 1. <https://www.ics.uci.edu/~eppstein/pubs/pubs.ff>
Score: 16.7520
- 2. https://www.ics.uci.edu/~gmark/Home_page/Publications.html
Score: 16.6795
- 3. <https://www.ics.uci.edu/~eppstein/pubs/all.html>
Score: 16.5286
- 4. <https://www.ics.uci.edu/~kay/courses/131/s04readings.html>
Score: 16.4300
- 5. <https://www.ics.uci.edu/~eppstein/bibs/eppstein.html>
Score: 16.1997

Enter query:

4. master of software engineering

Enter query: master of software engineering

Top results for 'master of software engineering' (36.9ms):

- 1. <https://www.ics.uci.edu/~kay/courses/h22/hw/DVD-random.txt>
Score: 34.0998
- 2. <https://www.ics.uci.edu/~kay/courses/h22/hw/DVD.txt>
Score: 34.0998
- 3. <https://www.informatics.uci.edu/grad/mswe/>
Score: 29.7632
- 4. <https://www.ics.uci.edu/~nenovita/vita.html>
Score: 29.1531
- 5. <https://mswe.ics.uci.edu/>
Score: 28.8621

Enter query: